

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Time:** 1 hour 45 minutes**Maximum Marks:** 25**General Instructions**

1. **All questions are compulsory.** Section A is done in **LibreOffice Calc**; Section B is done in **Scratch**.
2. **Suggested time:** 15 minutes entering the data in Calc, 30 minutes writing the formulas, chart and pivot table and emailing the file, and 60 minutes building the Scratch game.
3. Enter the two tables of Section A **exactly** as printed, on two sheets named **Product\_Master** and **Sales\_Data**. Columns F to I of **Sales\_Data** must be filled by **formula**, not typed by hand - typed values score no marks.
4. Save your spreadsheet as **YourName.ods** and email it to **shivam@cs17.org** and **hussain@cs17.org** before starting Section B.
5. Save your Scratch project as **YourName\_AppleCatcher** and leave it open on screen at the end of the examination.
6. Marks are shown in brackets against each part.

**Section A - Spreadsheet (Calc)****10 marks****Scenario**

**GreenLeaf Organics Pvt. Ltd.** is a retail chain selling organic products across four regions of the country - North, South, East and West. The company maintains a product master list (product code, name, category and unit price) on one sheet, and a sales register of every order received during January–February 2026 on another sheet. However, the sales register records **only the product code** - it does not repeat the product name, category or price.

The Sales Manager has asked you, the data assistant, to complete the sales register, calculate the revenue figures, and prepare a summary of regional and category-wise performance for the monthly review meeting.

Create a new spreadsheet with two sheets named exactly **Product\_Master** and **Sales\_Data**, and enter the two tables below.

**Sheet:** Product\_Master

|   | A            | B                  | C             | D          |
|---|--------------|--------------------|---------------|------------|
| 1 | Product Code | Product Name       | Category      | Unit Price |
| 2 | P001         | Green Tea Pack     | Beverages     | 250        |
| 3 | P002         | Cold Pressed Juice | Beverages     | 180        |
| 4 | P003         | Organic Honey      | Grocery       | 450        |
| 5 | P004         | Almond Butter      | Grocery       | 620        |
| 6 | P005         | Herbal Shampoo     | Personal Care | 320        |
| 7 | P006         | Aloe Face Wash     | Personal Care | 210        |
| 8 | P007         | Quinoa 1 kg        | Grocery       | 540        |
| 9 | P008         | Ginger Ale         | Beverages     | 150        |

**Sheet: Sales\_Data**

Columns **F**, **G**, **H** and **I** are shaded because they are to be left **blank** when you enter the data - you will fill them using formulas.

|    | A        | B          | C            | D      | E   | F            | G        | H          | I            |
|----|----------|------------|--------------|--------|-----|--------------|----------|------------|--------------|
| 1  | Order ID | Order Date | Product Code | Region | Qty | Product Name | Category | Unit Price | Total Amount |
| 2  | S01      | 05-01-2026 | P001         | North  | 12  |              |          |            |              |
| 3  | S02      | 07-01-2026 | P003         | South  | 8   |              |          |            |              |
| 4  | S03      | 09-01-2026 | P002         | North  | 20  |              |          |            |              |
| 5  | S04      | 11-01-2026 | P005         | East   | 15  |              |          |            |              |
| 6  | S05      | 14-01-2026 | P004         | West   | 6   |              |          |            |              |
| 7  | S06      | 16-01-2026 | P001         | South  | 10  |              |          |            |              |
| 8  | S07      | 18-01-2026 | P006         | North  | 25  |              |          |            |              |
| 9  | S08      | 21-01-2026 | P007         | East   | 9   |              |          |            |              |
| 10 | S09      | 23-01-2026 | P008         | North  | 30  |              |          |            |              |
| 11 | S10      | 25-01-2026 | P003         | West   | 14  |              |          |            |              |
| 12 | S11      | 28-01-2026 | P002         | South  | 18  |              |          |            |              |
| 13 | S12      | 30-01-2026 | P005         | North  | 11  |              |          |            |              |
| 14 | S13      | 02-02-2026 | P007         | South  | 7   |              |          |            |              |
| 15 | S14      | 04-02-2026 | P004         | East   | 5   |              |          |            |              |
| 16 | S15      | 06-02-2026 | P006         | West   | 22  |              |          |            |              |

**Question 1 - Completing and summarising the sales register****[10 marks]****1 (a) Looking up the product details****[2 marks]**

The sales register stores only the product code. On the sheet **Sales\_Data**, use the **XLOOKUP** function to bring in the missing details from the sheet **Product\_Master** for every order in rows 2 to 16:

- (i) In column **F** - the **Product Name** corresponding to the product code in column **C**.
- (ii) In column **G** - the **Category** of that product.
- (iii) In column **H** - the **Unit Price** of that product.

**Condition:** the formula must be written once in row 2 and copied down to row 16.

**1 (b) Order amounts and totals****[2 marks]**

- (i) In column **I** (Total Amount), write a formula to calculate the amount of each order as **Quantity × Unit Price**.
- (ii) In cell **I18**, calculate the **grand total** of all 15 orders.
- (iii) In cell **I19**, calculate the **average order value**.
- (iv) In cell **I20**, display the value of the **highest single order**.

Format the values in column **I** and in cells **I18:I20** as **Currency (Rs.)** with 2 decimal places.

**1 (c) Category and region figures****[2 marks]**

Prepare the answers in the cells indicated, using the **SUMIFS** function only:

- (i) Total sales amount of the **Beverages** category in the **North** region - cell **L3**

- (ii) Total sales amount of the **Grocery** category in the **West** region - cell **L4**
- (iii) Total sales amount of orders in the **East** region where Product Name is **Herbal Shampoo** - cell **L5**

Enter a suitable label for each value in the corresponding cells - **K3**, **K4** and **K5**.

#### 1 (d) Region summary and chart [2 marks]

- (i) In the range **K8:L12**, build a small summary table titled “Region-wise Sales”, listing the four regions in **K9:K12** and their total sales amounts in **L9:L12**. The totals must be calculated using the **SUMIF** function.
- (ii) Using this summary table, insert a **column (bar) chart** on the same sheet.

The chart must have - chart title: “Region-wise Sales - Jan-Feb 2026”; X-axis title: “Region”; Y-axis title: “Total Sales (Rs.)”; and **data labels** displayed on each column.

#### 1 (e) Pivot table summary [2 marks]

Insert a **pivot table** (Insert → Pivot Table) using the range **A1:I16** of **Sales\_Data** as the source data. Place the pivot table on a **new sheet** named **Pivot\_Summary**, with the following layout:

- (i) **Row fields** - Region
- (ii) **Column fields** - Category
- (iii) **Data / Values** - Sum of Total Amount

The pivot table must display **row totals** and **column totals**. Finally, sort the pivot table so that the region with the **highest** total sales appears first, and state that region’s name in a cell below the pivot table.

- End of Section A -

## Section B - Scratch Programming

15 marks

### Scenario

Your school is organising a **Health and Nutrition Week**. The organisers want a small computer game for the exhibition stall that encourages children to collect healthy fruit and avoid the garden pests.

You have been asked to build this game in Scratch. It is called **Apple Catcher**. A bowl at the bottom of the stage is moved left and right by the player. Apples fall from the top of the stage at random positions - catching an apple earns a point. Ladybugs also fall - touching a ladybug costs a life. The game runs for a fixed time or until all lives are lost, after which the final score is displayed.

### Question 2 - Design a game [15 marks]

#### 2 (a) Setting up the project [3 marks]

- (i) Choose a suitable **backdrop** for the stage (for example a garden, park or sky).
- (ii) Create **three sprites** and rename them exactly as **Bowl**, **Apple** and **Ladybug2**. Resize each sprite so that it is clearly visible but not larger than about one-fifth of the stage.
- (iii) Create **three variables** named **Score**, **Lives** and **Time**, and make all three visible on the stage.

- (iv) When the green flag is clicked, the game must **initialise** the variables to Score = 0, Lives = 3, Time = 60.

## 2 (b) Controlling the bowl

[3 marks]

Write the script for the `Bowl` sprite so that:

- (i) It starts at the **bottom centre** of the stage every time the game begins.
- (ii) Pressing the **right arrow** key moves it to the right, and the **left arrow** key moves it to the left. The movement must be **continuous** while the key is held down.
- (iii) The bowl must **not disappear off the edge** of the stage.

## 2 (c) Falling apples and scoring

[4 marks]

Write the script for the `Apple` sprite so that:

- (i) It appears at the **top** of the stage at a **random horizontal position**, using the `pick random` block.
- (ii) It **falls downwards** continuously.
- (iii) If it **touches the Bowl**, the score increases by 1, a **sound** is played, and the apple returns to the top at a new random position.
- (iv) If it reaches the **bottom** of the stage without being caught, it simply returns to the top at a new random position.

## 2 (d) The ladybug and losing lives

[3 marks]

Write the script for the `Ladybug2` sprite so that:

- (i) It behaves like the apple - falling from a random position at the top - but moves at a **noticeably different speed**.
- (ii) If it touches the `Bowl`, Lives decreases by 1 and a **different sound** is played.
- (iii) When Lives reaches 0, the message `Game Over` is **broadcast**.

## 2 (e) Timer and game over

[2 marks]

- (i) Write a script that reduces `Time` by 1 every second. When Time reaches 0, the message `Game Over` must be broadcast.
- (ii) On receiving `Game Over`: the Apple and the Ladybug2 must **hide**, a message showing the **final score** must be displayed on the stage (for example “Game Over! Your score is 12”), and the whole program must then **stop**.

- End of Section B -

- End of Question Paper -